

NERVES AND NEURONS

◆ A nerve impulse (signal) takes just **one hundredth of a second** to travel from the spinal cord to your big toe.

◆ The longest neurons—between spinal cord and big toes—are **3¼ ft (1 m) long**. They are also the longest cells in the body. The shortest neurons are $\frac{1}{100}$ in (1 mm) long.

◆ A neuron can transmit **1,000 nerve impulses** every second.

◆ The widest nerve, the sciatic, is **¾ in (2 cm) wide**. It extends from the lower back to the foot.

◆ A withdrawal reflex—when a nerve signal passes through the spinal cord, not the brain—can pull the hand away from a dangerously hot object in just **30 thousandths of a second (30 milliseconds)**. If the signal went via the brain it would take 800 milliseconds.

◆ Stretched out, the body's nerves would extend for more than **93,000 miles (150,000 km)**—more than the distance covered by flying between London and New York 25 times.

SENSES

★ Our **hearing range decreases with age**, which is why young people hear higher-pitched sounds than older people.

★ The tongue's **taste buds** are replaced every week, but the nose's smell receptors last for a month.

★ Chiles “taste” hot because they contain a substance that triggers the **tongue's pain receptors**.

★ Fingers are among the **most sensitive areas** of the body. Each fingertip has about 100 touch receptors.

★ The eyes contain **70 percent of the body's sensory receptors**, making

sight our most important sense.

★ Eyelids blink about **9,400 times** a day, helping to keep our eyes clean.

GENES

• Humans have **46 gene-carrying chromosomes** inside the nucleus of each body cell.

• Stretched out, the DNA in one human cell would extend over **6½ ft (2 m)**.



The farthest object that we see without a telescope is the Andromeda galaxy, which is 2.5 million light-years away.